

Does Your Router Actually Route?

A Mechanism-Validity Protocol for Adaptive Per-Node GNNs

ERP • AI Research*

Technical Report • July 2026

Abstract

Many recent graph neural networks adapt their computation per node. Some route nodes to topology-specialized experts, some mix low-pass, high-pass, and identity channels with node-wise weights, and some assign each node its own propagation depth. The usual motivation is that nodes sit in different local homophily regimes, and the usual evidence is aggregate accuracy on standard heterophily benchmarks. We argue that aggregate accuracy cannot establish that the adaptive mechanism itself is doing useful work, and we describe a protocol that can. The protocol compares the method’s LABEL-FREE router against a label-informed ORACLE router, a usage-matched RANDOM router, and a UNIFORM mixture with no routing, all parameter-identical by construction and asserted so at run time, using paired multisplit statistics. It is calibrated on preregistered positive-control graphs with planted per-node regimes. Applying it yields a conditional picture rather than a blanket verdict. On the small WebKB datasets the homophily-bucket oracle finds no exploitable signal for a topology-expert mixture (all three oracle-minus-uniform contrasts are nulls, robust across oracle settings), while the same graphs carry +4.1 to +7.1 pp of oracle signal for ACM-style channel mixing. On modern corrected benchmarks the same topology-expert model shows significant oracle headroom on eight of nine datasets (+0.8 to +38.8 points, one null), of which the label-free router realizes between none and half, exceeding the homophily oracle on one dataset; frozen-model interventions nevertheless show the learned assignment beats its own node-shuffled gates on every dataset, separating assignment value from training effects. On the planted control the oracle gain (+26.2 pp over the strongest single expert) decays monotonically to a preregistered measured null as regime variance is diluted, and the same control detects planted signal for a foreign architecture (+35.9 pp for ACM-GNN). We also show that a positive control must plant the signal the mechanism claims to exploit; a depth-routing method is correctly not helped by a filter-planted control (−9.3 pp). Between drafts, an external code-level review and the integrity system built in response corrected a capacity confound, a generator confound, and

silent data contamination in this paper’s own evidence; every correction is released, and none changed a verdict that survives in this version. The protocol, the generators, more than 15,000 runs, and a machine-checkable ledger binding each claim to the released data are public, and we suggest the accompanying checklist as a standard report for adaptive-GNN papers.

1 Introduction

Message passing aggregates a node’s neighborhood into its representation. Under homophily this sharpens class information; under heterophily it can destroy it. This observation has produced a family of per-node adaptive architectures: mixtures of topology-specialized experts, node-wise channel mixing between aggregation, diversification, and identity filters (Luan et al., 2022), and node-specific propagation depths (Hevapathige et al., 2026; Wang et al., 2024b; Abbahaddou et al., 2025). The motivation is reasonable. Graphs are not uniformly homophilic, so a single global propagation rule should be suboptimal, and a router that treats each node according to its local regime should help.

The evidence offered for these methods is almost always aggregate accuracy on standard heterophily benchmarks, and this is a weak instrument for two reasons. The first is that the classic benchmarks are tiny. Cornell and Texas contain 183 nodes and, under the standard splits, 37 test nodes, so a single test node is worth 2.7 accuracy points. Evaluation fragility of this kind has been documented in general (Shchur et al., 2018), and Platonov et al. (2023) found specific defects in the heterophily suites, including duplicate-node leakage in Squirrel and Chameleon. The second reason has received less attention: even a genuine accuracy gain does not show that the router caused it. A routed mixture differs from its baselines in capacity, in architecture, and in training dynamics. The router is only one ingredient.

We therefore ask a narrow question: when a per-node adaptive GNN appears to help, is the router actually doing useful work? Our answer is a protocol rather than a verdict. The central tool is an ORACLE router, a router given label-derived information about each node’s local regime. This is a deliberately favorable condition. If even the oracle cannot beat a UNIFORM mixture of the

*Correspondence: ssh@erp.ai.

same experts with no routing at all, then that architecture has no routing signal to find on that benchmark, and any accuracy gain of the full model must come from something else. Conversely, when the oracle does beat uniform, the gap measures the routing headroom against which a label-free router can be scored.

Two properties make the protocol trustworthy. It is calibrated: on generator-controlled graphs with planted per-node regimes, the oracle gain is large (+26.2 pp against the strongest single expert) and decays monotonically to a measured null as the regime variance is diluted away (Fig. 1), on a preregistered replication with held-out generator seeds, so the instrument detects signal exactly when signal exists. It is also portable: the same planted control detects signal for a published foreign architecture (ACM-GNN, +35.9 pp), so the calibration is not an artifact of our own model family.

Applying the protocol produced results we did not predict. We expected a clean negative result, since our own topology-expert mixture’s WebKB gains had already failed the oracle test. Instead we found a conditional landscape (Table 1). The same WebKB graphs that contain no signal for topology-expert routing contain +4 to +7 pp of oracle signal for channel mixing. The same topology-expert model that fails on WebKB has +0.8 to +38.8 points of oracle headroom on modern corrected benchmarks, of which its label-free router realizes between none and half, and on one dataset more than all of it. Whether per-node adaptation has exploitable signal is a property of the (architecture, benchmark) pair, and aggregate accuracy cannot tell these cases apart. The controls can.

1.1 Contributions

1. **A mechanism-validity protocol** for per-node adaptive GNNs (§3): oracle, uniform, random, and label-free router conditions, all parameter-identical, single-expert controls, paired statistics, and router diagnostics, packaged as a reusable checklist (Table 2).
2. **Calibrated positive controls** (§4): a planted-regime generator whose oracle gain decays monotonically to a measured null under regime dilution, validated on our architecture and on a published one, together with the principle that controls must be mechanism-matched. A filter-planted control is correctly unhelpful for a depth-routing method (§4.4).
3. **A conditional empirical landscape** (§5–6): no topology-expert routing signal on WebKB even for an oracle; significant channel-mixing oracle signal on the same graphs; large oracle headroom on modern corrected benchmarks with partial label-free recovery; and the oracle-to-label-free gap as a measure of the headroom that remains for router research.

4. **An evaluation-granularity analysis** (§7): minimum detectable effects for the paired protocol on each benchmark, showing that the WebKB classics cannot resolve the effect sizes typically claimed on them.

1.2 Scope and non-claims

We do not claim that adaptive per-node GNNs fail in general; several of our own results show large routing headroom. We do not claim that the WebKB datasets are unusable, only that they cannot validate per-node routing mechanisms without the controls given here. We do not claim that our reimplementation of any published method matches its authors’ tuned results. The protocol compares router conditions within a fixed architecture, which is what makes it insensitive to absolute tuning. Finally, the oracle is an upper-bound condition by construction. Where it nearly saturates a metric (§6) we note this and rely on the label-free numbers instead.

2 Related Work

Evaluation pitfalls and benchmark critique. Shchur et al. (2018) showed that GNN rankings are fragile to splits and tuning. Platonov et al. (2023) audited the heterophily suites specifically, identified duplicate-node leakage in Squirrel and Chameleon, and introduced corrected datasets on which standard GNNs are often strong; subsequent work questions whether heterophily-specific architectures and homophily metrics deliver what they claim under fair tuning (Luan et al., 2024). Our work is downstream of these. They audit datasets and tuning; we audit mechanisms. The protocol here is what a method paper would run after choosing sound benchmarks.

Heterophily architectures. H2GCN (Zhu et al., 2020) separates ego and neighbor embeddings, GPRGNN and FAGCN learn global filter responses, LINKX (Lim et al., 2021) is a strong propagation-light baseline, and signed-message models (Yan et al., 2022) reweight messages rather than route nodes. These works motivate strong baselines but are not per-node adaptive in our sense.

Per-node adaptive methods. ACM-GNN (Luan et al., 2022) mixes aggregation, diversification, and identity channels node-wise; it is the closest published relative of topology-expert routing and one of our case studies. Adaptive-depth methods select per-node propagation depth: AD-GNN (Hevopathige et al., 2026) formalizes a signal-preservation factor and assigns node-specific depths, the Heterophilic Snowflake hypothesis (Wang et al., 2024b) argues that each node has an optimal receptive field, and ADMP-GNN (Abbahaddou et al., 2025)

Table 1: The findings matrix. ORACLE—UNIFORM measures whether routing signal exists for a mechanism on a benchmark family (decision rule D1, §3.3); label-free (LF) recovery is the fraction of that gap the method’s own router realizes. Details: WebKB §5, planted controls §4, modern benchmarks §6.

Method	Benchmark family	ORACLE—UNIFORM	LF recovery	Verdict
TopoMoE	WebKB classics	−0.1 to +1.6 pp (all ns)	—	no signal found by this oracle class
ACM-GNN	WebKB classics	+4.1 to +6.9 pp	0.34–0.90	channel-mixing signal present
AD-GNN	WebKB classics	−3.7 to −1.0 pp	—	depth oracle unhelped
TopoMoE	modern corrected (9)	−0.9 to +38.8	0–0.5 ^a	headroom on 8/9; partial recovery
TopoMoE	planted PC-1c-R	+12.0 pp (+26.2 vs best single)	0.94	preregistered calibration passes
ACM-GNN	planted PC-1b	+35.9 pp	0.72	calibration passes (foreign arch.)
AD-GNN	planted PC-1b (filter)	−9.3 pp	—	mechanism-mismatched control

^aOn roman-empire the label-free router exceeds the homophily oracle (§6).

and curvature-based depth selection (Hevopathige et al., 2025) are further variants. Wang et al. (2024) give theory for when heterophily hurts as a function of neighborhood distributions and degree. None of these works reports an oracle-versus-uniform control. Supplying that control, and the calibrated instrument needed to interpret it, is the contribution of this paper.

Routing analyses outside GNNs. Related controls exist in other communities. In sparse mixture-of-experts language models, Roller et al. (2021) showed that fixed random hashing can match learned routing, and router-collapse analyses are standard. Dikkala et al. (2023) compare learned against non-trainable routing in MoE Transformers and show theoretically and empirically that routers recover planted cluster structure in synthetic data; our planted-regime controls share that logic, transported to per-node routing on graphs and extended with the oracle, capacity, and calibration machinery. Early-exit networks report oracle exit selection as an upper bound (Teerapittayanon et al., 2016; Boenke et al., 2025). Our protocol differs in how the oracle is used: not as an upper bound to aspire to, but as a falsification test (if oracle $\leq$ uniform, no signal of the oracle’s class is detectable for the mechanism), calibrated against planted-regime controls and run with parameter-identical conditions on graph benchmarks.

3 The Mechanism-Validity Protocol

3.1 Setting and notation

Let $G = (V, E)$ be a graph with node features $\mathbf{x}_v$, labels y_v , and neighborhoods $N(v)$. Given community labels $c(\cdot)$ from unsupervised detection (Louvain or Leiden), the boundary score of a node is

$$\beta(v) = \frac{|\{u \in N(v) : c(u) \neq c(v)\}|}{|N(v)|}, \quad \varphi(v) = 1 - \beta(v), \quad (1)$$

where $\varphi(v)$ is the same-community neighbor fraction. We write local label homophily as $h(v) = |\{u \in N(v) : y_u = y_v\}|/|N(v)|$ and local feature homophily as $h_x(v) = \frac{1}{|N(v)|} \sum_{u \in N(v)} \cos(\mathbf{x}_v, \mathbf{x}_u)$; the latter needs no labels. A per-node routing policy is a map $\pi(v) = g(s(v))$ from a signal field $s : V \rightarrow [0, 1]$ to experts, channels, or depths; the UNIFORM condition below is the constant policy.

For the analysis in §4 and §6 we also define a community-aware redundancy field over the same-community neighbors $N_s(v) = \{u \in N(v) : c(u) = c(v)\}$:

$$R_c(v) = \underbrace{\left(|N_s(v)|\right)^{-1} \sum_{\substack{u, w \in N_s(v) \\ u < w}} \cos(\mathbf{x}_u, \mathbf{x}_w)}_{\text{intra-community coherence}} \times \varphi(v). \quad (2)$$

When $|N_s(v)| < 2$ the coherence term is undefined and we set $R_c(v) = 0$ by convention; isolated nodes take $h_x(v) = 0$ likewise. Unlike homophily, which compares each neighbor to the node itself, R_c measures how coherent the neighbors are with one another within the node’s own community. The two quantities are not interchangeable, in the following exact sense.

Proposition 1 (Separation). *There exist a graph, community labels, and two nodes v_1, v_2 with identical degree and identical local feature homophily h_x , but different R_c . Consequently, R_c cannot be written as any function of degree and h_x .*

Proof. Take two disjoint 4-stars in one graph, with community labels given by connected components, so $\varphi(v_1) = \varphi(v_2) = 1$ and both centers have degree 4. Features are unit vectors in $\mathbb{R}^3$. Center v_1 has $\mathbf{x}_{v_1} = (1, 0, 0)$ and all four leaves share the feature $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0)$: every leaf is at angle $\pi/4$ from v_1 , so $h_x(v_1) = \frac{\sqrt{2}}{2}$, and all pairwise leaf cosines equal 1, so $R_c(v_1) = 1$. Center v_2 has $\mathbf{x}_{v_2} = (1, 0, 0)$ and leaves $\mathbf{w}_i = (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \cos \alpha_i, \frac{\sqrt{2}}{2} \sin \alpha_i)$ with $\alpha_i \in \{0, \pi/2, \pi, 3\pi/2\}$: each leaf is again at angle $\pi/4$ from v_2 , so $h_x(v_2) = \frac{\sqrt{2}}{2}$, while $\langle \mathbf{w}_i, \mathbf{w}_j \rangle = \frac{1}{2} + \frac{1}{2} \cos(\alpha_i - \alpha_j)$ gives four pairs at $\frac{1}{2}$ and two pairs at 0, so $R_c(v_2) = \frac{1}{6}(4 \cdot \frac{1}{2} + 2 \cdot 0) = \frac{1}{3}$. The two centers agree on degree and h_x and differ on R_c . $\square$

Structural fields can therefore carry routing information that homophily alone does not; §6 shows the distinction is not merely formal. The label-free router evaluated in §5 and §6 uses the structural field $\{h_x, \text{spectral ratio, log-degree, clustering}\}$.

3.2 Router conditions

Fix an architecture with K experts (or channels, or depth levels) and a per-node assignment mechanism. All conditions below share the identical expert family, training procedure, and budget; only the assignment signal changes. In our instantiation all four conditions are parameter-identical to the digit: the UNIFORM condition is the same routed model with its gates frozen at exactly $(1/3, 1/3, 1/3)$, verified by a runtime identity audit on every batch. (An earlier version of this work used a smaller shared-stack model as UNIFORM, a capacity confound caught in external review; that model is retained as a secondary baseline in App. F, and the RANDOM condition, which carries the identical parameter count with an uninformative signal, independently controls capacity throughout.)

label-free. The method’s own router, learned or structural. This is the condition under evaluation.

oracle. Assignment from label-derived local-regime information. On planted graphs this is the true regime id; on real graphs it is a bucketing of the label homophily $h(v)$ of §3.1 (low-pass if $h(v) \geq 0.6$, high-pass if $h(v) \leq 0.4$, identity otherwise). For channel-mixing methods we use both a hard one-hot and a soft $(0.8, 0.1, 0.1)$ variant. The oracle is deliberately favorable: it answers whether there is any signal to route on, not whether this is the best possible router.

random. Fixed random assignment with the oracle’s usage histogram, so that the expert-usage budget matches. This is the floor for the label-free router.

uniform. Equal mixing of all experts, i.e. no routing: the routed model with its gates frozen. Together with single-expert baselines, this separates routing effects from training effects and identifies which expert carries a given dataset.

3.3 Decision rules

D1 (signal exists): ORACLE > UNIFORM. **D2** (the router finds it): LABEL-FREE > RANDOM. **D3** (not capacity): ORACLE > RANDOM at identical parameter count. **D4** (the instrument works): D1 must hold on a mechanism-matched positive control. A mechanism claim is supported only if D1 through D4 all hold. We flag one asymmetry here because our first draft got it wrong: a D1 failure is evidence about the oracle’s signal class, not about all possible routers. If D1 fails while D4 holds,

the benchmark contains no signal *detectable through that oracle’s signal* for that mechanism; we harden such verdicts by sweeping the oracle’s thresholds and mapping (§5), but a different signal class could in principle succeed, and our own results contain an example of an oracle understating attainable routing (§6). If D1 holds while D2 fails, signal exists but the router misses it.

Statistics. 10 splits $\times$ 3 seeds are run per condition. Seeds are averaged within each split as technical replicates, and all comparisons are paired over the resulting 10 per-split differences (Penn94: 5). The exact Wilcoxon signed-rank test is the declared primary test; the paired t is a sensitivity analysis, with Holm–Bonferroni correction within each comparison-type family across datasets (Holm, 1979), mean differences, 95% CIs, and the paired effect size $d_z = \bar{d}/s_d$. The exact Wilcoxon floor is $p = .00195$ at $n=10$ and $.0625$ at $n=5$, which is why the paired t is reported alongside. We additionally report each benchmark’s minimum detectable effect under this design (§7). As a negative control on the statistics themselves, we ran the identical paired machinery on A/A comparisons, testing each router condition against itself across seeds. The control fired: 12 of 387 comparisons survived Holm correction. The cause is real rather than a pipeline defect: a fixed seed reproduces the identical initialization across splits, and some initializations are consistently better, so seeds are technical replicates rather than independent samples. All inferential results in this paper therefore use the corrected primary analysis described above; the A/A record and the corrected re-analysis ship with the release (App. E).

4 Calibration: Planted-Regime Positive Controls

4.1 The PC generator

PC graphs have $N=10,000$ nodes, 5 classes, and roughly 50 contiguous patches, each assigned one of three locally coherent regimes (40/40/20): **L** (homophilic wiring; labels recoverable by low-pass aggregation), **H** (anti-correlated wiring in which class c links to a different class; labels recoverable by differencing), and **M** (random wiring; labels carried by features only). Local coherence matters here, because i.i.d. per-node regimes average out under aggregation and destroy the planted signal. A dilution parameter λ converts nodes to the feature-only regime, and $\lambda=1$ removes all per-node regime variance. The generator used for every result in this section, PC-1c, is deconfounded so that λ varies exactly one thing: feature noise is constant across the ladder, the converted node sets are nested (one permutation per graph seed), unconverted nodes retain their original wiring and features, and the M-regime feature noise is calibrated so that no arm saturates the metric (single-M expert accuracy near 0.80 on M-nodes), mak-

Table 2: The diagnostic checklist for per-node adaptive GNN claims.

Claim in a paper	Required control
Routing signal exists on this benchmark	ORACLE vs UNIFORM (D1)
The learned router finds it	LABEL-FREE vs RANDOM (D2)
The gain is not capacity	ORACLE vs RANDOM at identical size (D3)
Which expert carries performance	single-expert baselines
The router specializes (vs collapses)	usage entropy; MI vs shuffled null
The result is resolvable	paired multi-split tests; MDE (§7)
Depth adaptation is in a useful regime	two fixed-depth runs: $d_4/d_2 < 1$ (§9)
The instrument itself works	mechanism-matched positive control (D4)

Algorithm 1 The mechanism-validity protocol for a per-node adaptive GNN

Require: architecture with experts $E_1, \dots, E_K$ and router signal s ; benchmark G ; splits $\times$ seeds

- 1: Build parameter-identical conditions sharing experts, training, and budget: LABEL-FREE ($\pi = g(s)$), ORACLE (label-derived regime buckets), RANDOM (usage-matched), UNIFORM (gates frozen equal), plus single-expert baselines; assert identity at run time
- 2: Train and evaluate every condition on identical (split, seed) pairs; average seeds within splits as technical replicates
- 3: Paired exact Wilcoxon and t tests over splits; Holm per comparison family; effect sizes; CIs
- 4: **D4:** verify ORACLE > UNIFORM on a mechanism-matched planted control; abort if not (instrument invalid)
- 5: **D1:** ORACLE > UNIFORM on G ? **No** $\Rightarrow$ no exploitable signal for this mechanism on G
- 6: **D2:** LABEL-FREE > RANDOM? **No** $\Rightarrow$ signal exists but the router misses it
- 7: **D3:** ORACLE > RANDOM at identical parameter count? **No** $\Rightarrow$ gain is capacity or training dynamics, not routing
- 8: Report the findings-matrix row: ORACLE–UNIFORM, label-free recovery, verdict

ing the $\lambda=1$ endpoint a measured value rather than a ceiling artifact.¹ Every graph is checked before training by automated instrument gates: exact wiring fractions per regime, planted-signal recovery probes, nesting and single-variable invariants.

The generator admits a closed-form reading in the contextual stochastic-block-model setting.

Proposition 2 (Redundancy on a planted model). *Let node v have neighbor features $\mathbf{x}_u = \boldsymbol{\mu}_{c(u)} + \mathbf{z}_u$ with $\mathbf{z}_u \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D)$ mutually independent and $\|\boldsymbol{\mu}_c\| = \Delta$ for every*

¹Two earlier versions of this control are retained for the record: PC-1 saturates at $\lambda=1$ (every arm reaches accuracy 1.0), and both PC-1 and PC-1b coupled feature noise to λ , a single-variable violation caught in external review. App. A and App. F document both, together with a data-integrity defect found in the historical PC-1b runs.

community c . Then, to leading order as $D \rightarrow \infty$,

$$\mathbb{E}[R_c(v)] \approx (1 - \beta(v)) \frac{\Delta^2}{\Delta^2 + D}, \quad (3)$$

strictly decreasing in the boundary score $\beta(v)$.

Proof. For same-community neighbors u, w of v , $\langle \mathbf{x}_u, \mathbf{x}_w \rangle = \Delta^2 + \boldsymbol{\mu}^\top (\mathbf{z}_u + \mathbf{z}_w) + \langle \mathbf{z}_u, \mathbf{z}_w \rangle$, with expectation Δ^2 and standard deviation $O(\sqrt{\Delta^2 + D})$, while $\|\mathbf{x}_u\|^2$ concentrates at $\Delta^2 + D$ with $O(\sqrt{D})$ fluctuations. The cosine of each same-community pair therefore concentrates at $\Delta^2/(\Delta^2 + D)$ with $O(D^{-1/2})$ absolute error, and by linearity so does the intra-community coherence term of Eq. (2). (For fixed Δ the mean itself is $\Theta(D^{-1})$, so the relative error grows; the approximation in Eq. (3) is sharp in the regime $\Delta^2 = \Theta(D)$ and is an absolute-error statement otherwise.) The factor $\varphi(v) = 1 - \beta(v)$ is deterministic given the partition, and cross-community neighbors do not enter the coherence term, giving the product form. $\square$

Planted regime structure therefore induces a structural field that a label-free router can read, at a rate set by the class-mean norm relative to the ambient dimension. In these terms, the PC-1b calibration is a reduction of Δ on the M-patches, and the dilution parameter λ flattens the field by making a growing fraction of nodes share one regime.

Sanity checks. On the held-out evaluation seeds, each single expert dominates its own regime (single-L on L-nodes 0.815; single-H on H-nodes 0.680; single-M on M-nodes 0.809, inside its preregistered calibration band [0.74, 0.86]), and all conditions are parameter-identical by construction, so the gain to be measured is not a capacity effect.

4.2 Result: the dilution curve, under two contrasts

The two contrasts in Table 3 answer different questions, and the gap between them turned out to matter. At $\lambda=1$ every node is M, so the oracle degenerates to a constant all-M policy; the usage-matched RANDOM condition coincides with the same constant policy there, and a constant policy is not per-node routing. The per-node measure,

Table 3: PC-1c-R, the preregistered replication on held-out generator seeds (one graph per seed, $n=10$ paired graphs per point; Holm–Wilcoxon). The per-node routing measure ORACLE–best-single decays monotonically to a measured null; ORACLE–UNIFORM terminates at a small positive residual that is not per-node routing (see text).

λ	0.00	0.25	0.50	0.75	1.00
ORACLE–single-M (pp)	+26.17	+19.71	+13.66	+7.09	+0.09
95% CI	[25.3, 27.0]	[19.2, 20.2]	[13.1, 14.2]	[6.8, 7.4]	[−0.16, +0.35]
p_{Holm}	.0098	.0098	.0098	.0098	0.43
ORACLE–UNIFORM (pp)	+12.03	+14.13	+13.93	+7.48	+0.56
95% CI	[11.1, 13.0]	[13.4, 14.9]	[13.5, 14.4]	[7.0, 7.9]	[+0.34, +0.79]
p_{Holm}	.0098	.0098	.0098	.0098	.0098

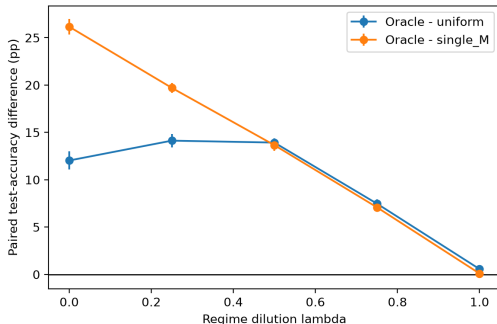


Figure 1: Per-node routing gain decays with planted regime variance. On PC-1c-R, ORACLE–best-single decays monotonically from +26.2 pp to a measured null at $\lambda=1$ (CI crossing zero, no arm at a ceiling), while ORACLE–UNIFORM terminates at a small positive residual that the text decomposes as expert-composition, not routing ($\pm 95\%$ CI bars; values and tests in Table 3). The WebKB results of §5 sit at the $\lambda \rightarrow 1$ end of this curve.

ORACLE–best-single-expert, therefore terminates at an exact measured null (+0.09 pp, CI [−0.16, +0.35]), while ORACLE–UNIFORM retains a small significant residual (+0.56 pp): the equal mixture averages in a low-pass expert that scores at chance on wiring-free graphs, a homogeneous expert-composition cost, not a routing signal. An earlier run of this control fired its preregistered endpoint gate on exactly this residual; the decomposition above is what the investigation established, and the replication’s corrected endpoint null was preregistered before launch and passed on ten held-out generator seeds (App. F). The ORACLE–UNIFORM curve is also not monotone near small λ , because conversion changes the node-type mix as well as the heterogeneity; we report the curve as measured and claim monotone decay only for the per-node contrast.

The label-free structural router recovers 0.94 of the oracle-over-uniform gain at $\lambda=0$ (+11.29 pp against +12.03; accuracies .7552, .7626, .6423 for LABEL-FREE, ORACLE, UNIFORM). When regime signal exists, a label-free router can find it. The clean rerun of the legacy-profile PC-1b ladder agrees across generators (+12.06 pp at $\lambda=0$ against PC-1c-R’s +12.03; endpoint residual

+0.46 against +0.56), so the calibration is not an artifact of one generator implementation. On the strong-signal planted profile the label-free router’s dominant assignment carries near-total mutual information with local homophily (normalized MI effect 0.974 above a 200-permutation shuffled null, permutation $p=.005$); diagnostics for the real datasets appear in §8.

4.3 Portability: a foreign architecture

We replaced ACM-GNN’s learned node-wise channel weights with the same four conditions. On PC-1b the hard oracle gains +35.89 pp over uniform channel mixing ($p_{\text{Holm}}=.0098$), and ACM’s own learned (label-free) mixing gains +25.92, recovering 0.72 of the oracle gap. The instrument detects planted signal for an architecture it was not designed around.

4.4 Mechanism-matched controls: the AD-GNN case

AD-GNN (Hevathige et al., 2026) routes depth, not filter type. On the filter-planted PC-1b its depth oracle is negative (−9.27 pp, $p_{\text{Holm}}=.0098$). There is no depth-relevant signal in filter-type regimes, and forcing depth assignments by an irrelevant partition hurts. This is not a failure of the instrument. It is a design principle that we propose as part of the protocol: a positive control must plant the signal that the mechanism claims to exploit. Filter-planted controls calibrate filter and channel routing. Depth-routing methods require a depth-planted control, in which labels are recoverable at hop 1 in some regions and only at hop 3 in others; we leave this construction as the natural extension. Accordingly, our AD-GNN results on real data (§5) are reported without a planted calibration and are scoped as such.

5 Case Studies on the WebKB Classics

5.1 TopoMoE: the motivating failure

TopoMoE dispatches nodes to low-pass, high-pass, and MLP experts via a label-free structural router. Its development history, reported in full in App. B, is instructive. The model significantly beat a gated multi-head GAT control on Cornell, Texas, and Wisconsin (+17 to +23 pp), but that control turns out to be a bottom-quartile model on exactly those datasets, and against a strong baseline matrix TopoMoE was best on one of five heterophilic datasets. The router-quality experiment explains why the WebKB wins were never the router’s doing:

By decision rule D1, no signal is found for this mechanism on these graphs through the homophily-bucket oracle class: all three oracle-versus-uniform contrasts are nulls at $p_{\text{Holm}}=1.0$, and the verdict is robust to the oracle’s settings, since rerunning the oracle arm with thresholds (0.30, 0.70) and (0.45, 0.55) and with soft assignment turns no WebKB dataset significantly positive against uniform (largest point estimate +0.54 pp, all $p_{\text{Holm}} \geq .18$). The router does beat its usage-matched random floor everywhere (+6.0 to +7.2 pp, $p=.006$), so D2 passes vacuously against a floor that D1 renders moot. Combined with §4, where the same router recovers 0.94 of a large oracle gain, the conclusion is specific: the router is not broken. These benchmarks contain no per-node signal that this expert family can exploit through any regime signal we tested; per §3.3, a different signal class could in principle succeed, and finding one would itself be a result. Table 4 was re-derived from scratch in the audited harness after an external review; the originally imported runs’ conclusions reproduce exactly, with point-estimate drift documented in App. F.

5.2 ACM-GNN: the same graphs are not empty for a different mechanism

ACM-GNN’s soft oracle beats uniform channel mixing on Cornell and Texas under both tests, and on Wisconsin by the t-test only (Table 5). Its learned mixing recovers 0.90 of the oracle gain on Cornell but only 0.34 and 0.43 on Texas and Wisconsin. Together with §5.1, this rules out the tempting general reading of the TopoMoE result. WebKB is not signal-free. Signal existence is a property of the (mechanism, benchmark) pair: selecting among aggregation, diversification, and identity channels has per-node signal on these graphs that selecting among low-pass, high-pass, and MLP experts does not. No aggregate-accuracy comparison between the two architectures could reveal this distinction. The oracle and uniform controls do.

6 Modern Corrected Benchmarks: Headroom and Recovery

On the corrected Platonov suite (Platonov et al., 2023), Penn94 (Lim et al., 2021), and Actor, the results reverse (Table 6). The same topology-expert model that had no oracle signal on WebKB shows significant oracle headroom on eight of nine datasets against a parameter-identical uniform, from +0.8 to +38.8 points. Its label-free router recovers about half of that headroom on Penn94 and amazon-ratings, smaller fractions on the AUC datasets, and nothing significant on three. Squirrel-filtered is a null (−0.91 pp, $p_{\text{Holm}}=.13$): even after Platonov filtering, this graph offers no detectable routing signal for the low-pass/high-pass/identity family.

An earlier version of this table, produced before an external review, used a smaller shared-stack model as UNIFORM and consequently overstated both the headroom and the recovery; the most instructive casualty of that correction is minesweeper, where an apparent +9.4 AUC-point label-free gain over the undersized uniform becomes −0.41 against the parameter-identical one. The gain was model size, not routing, and the protocol’s own capacity rule (D3), properly implemented, is what exposed it (App. F).

Two caveats apply. First, on roman-empire the label-free structural router exceeds the homophily-bucket oracle (+10.7 against +6.4 pp at identical capacity). The narrow reading is that the coarse homophily bucketing is not the optimal oracle on that graph. An oracle condition lower-bounds the best router given its particular signal, and a better structural signal can beat a worse label-derived one. Proposition 1 shows formally that partition-aware structural fields can carry information no function of degree and homophily encodes; the roman-empire result is consistent with that mechanism, though we have not run the ablation that would attribute it (the router’s field does not include R_c itself; see §10). Second, on minesweeper and questions the oracle nearly saturates AUC (.998 and .995; tolokors reaches .920). On binary graphs, knowing a node’s true local homophily together with its neighborhood is strongly informative about its label, so the oracle gaps on all three † datasets are generous upper bounds; the significant label-free gains (+1.5 and +6.9 AUC points on tolokors and questions) are the more meaningful quantity. Absolute accuracies throughout Table 6 are below tuned state of the art on these datasets because the backbone is the small fixed protocol model; the within-architecture differences between router conditions are the object of study (§1.2).

Capacity is controlled twice in Table 6: the UNIFORM condition is parameter-identical by construction, and the RANDOM condition provides an independent check, since oracle-minus-random at identical capacity is significant and positive on eight of nine datasets (+0.84 to

Table 4: TopoMoE router grid on WebKB (accuracy; split-level design, $n=10$; audited clean rerun replacing the originally imported runs, whose conclusions it reproduces, App. F). The oracle cannot beat no-routing on any of the three datasets.

Dataset	LABEL-FREE	ORACLE	RANDOM	UNIFORM	ORACLE-UNIFORM		
					diff (pp)	95% CI	p_{Holm}
Cornell	.6559	.6586	.5865	.6423	+1.62	$[-2.6, +5.8]$	1.0
Texas	.7784	.7586	.7180	.7595	-0.09	$[-3.0, +2.8]$	1.0
Wisconsin	.7418	.7712	.6699	.7614	+0.98	$[-1.9, +3.9]$	1.0

Table 5: Published-method router grids on WebKB and the planted control (pp vs. UNIFORM; Holm-Wilcoxon p in parentheses, split-level design; all rows from the audited clean rerun, which reproduces the accepted WebKB group means of the original runs to four decimals, App. F). ACM-GNN’s channel-mixing oracle finds signal where TopoMoE’s topology experts find none. AD-GNN’s depth oracle is unhelped, and it lacks a mechanism-matched planted control (§4.4).

Method	Condition	Cornell	Texas	Wisconsin
ACM-GNN	ORACLE (soft)	+6.58 (.012)	+6.94 (.023)	+4.05 (.055)
	ORACLE (hard)	+4.14 (.010)	+7.12 (.010)	+2.75 (.26)
	LABEL-FREE	+5.95 (.082)	+2.34 (.39)	+1.76 (.39)
AD-GNN (reimpl.)	ORACLE	-3.06 (.18)	-0.99 (.34)	-3.66 (.10)
	LABEL-FREE	-3.06 (.18)	-1.80 (.39)	-2.81 (.39)
PC-1b planted control (ORACLE, hard)				
ACM-GNN			+35.89 (.010)	
AD-GNN		-9.27 (.010), mechanism-mismatched control		

+33.86 pp, ten of ten splits positive on each) and null on squirrel-filtered, exactly where no signal exists.

6.1 Assignment value at fixed experts: frozen-model interventions

The retrained comparisons above measure a total system effect: routing, expert specialization under that routing, and optimization all change together. To isolate the inference-time assignment itself, we take each trained LABEL-FREE model, freeze it, and re-evaluate the SAME weights under five gate settings: the learned gates, node-shuffled learned gates (permuting the gate vectors across nodes, which preserves the full gate distribution and softness and destroys only the node-to-gate correspondence; 20 permutations), the global-mean gate, equal gates, and oracle one-hot gates. Learned-minus-shuffled is then a pure test of whether WHERE the gates point matters, at fixed experts.

It does, on all fourteen planned dataset groups (full coverage; six groups were re-based or recovered during the audit, App. F). The learned assignment beats its own shuffled counterpart everywhere it should: +1.81 pp on Actor, +2.50 on amazon-ratings, +4.55 on chameleon-filtered, +2.98 on minesweeper, +3.89 on squirrel-filtered, +11.11 on tolokera, +15.96 on questions, +29.47 on roman-empire, +7.90 on Penn94 (by the paired t at $n=5$), and +25.52 on the planted control at $\lambda=0$, all significant under the primary test except Penn94 as

noted. Across the planted dilution ladder the intervention traces a second, independent decay curve at fixed experts: $+25.52 \rightarrow +26.88 \rightarrow +21.52 \rightarrow +9.07$ and then -0.00 pp (CI $[-0.10, +0.09]$) at $\lambda=1$, an exact null exactly where per-node assignment has nothing left to do. Two readings deserve note. Squirrel-filtered, a D1 null in Table 6, nevertheless shows +3.9 pp of assignment value at fixed experts: retraining under different assignments washes the signal out (the total-system effect is null), yet the trained experts are genuinely heterogeneous per node. And minesweeper, whose label-free router does not beat uniform end to end, still loses 3.0 AUC points when its gates are shuffled, so the router is doing per-node work even where the system-level contrast cannot show it. The protocol needs both measurements, and they answer different questions.

Taken together, on modern benchmarks the gap between oracle and label-free routing is measured, unrealized headroom for router research: 2.2 pp on Penn94, 9.6 AUC points on minesweeper, and effectively all of the headroom on chameleon-filtered and Actor, where the label-free router recovers nothing significant. Per-node routing on these benchmarks is neither solved nor empty. It is an open problem, and the protocol supplies the measurement.

Table 6: TopoMoE router grid on modern benchmarks, with the parameter-identical equal-gate UNIFORM. The metric follows each dataset’s standard protocol: ROC-AUC for the imbalanced binary datasets ($\dagger$; majority-class row = 0.5), accuracy otherwise. All significant ORACLE–UNIFORM differences have $p_{\text{Holm}} \leq .028$ (primary Wilcoxon; Penn94 by paired t , its 5-split Wilcoxon floor being .0625); squirrel-filtered is a null ($p_{\text{Holm}} = .13$). Non-significant LABEL-FREE cells are marked ns; minesweeper’s is significantly negative. On the $\dagger$ datasets the label-derived oracle is especially informative, and on minesweeper and questions it nearly saturates AUC; those oracle columns should be read as generous upper bounds, with the LABEL-FREE columns carrying the practical comparison. Two questions/UNIFORM runs collapse to constant prediction (AUC 0.5), a documented instability retained in the data.

Dataset	UNIFORM	ORACLE	LABEL-FREE	orc-unif	LF-unif	LF rec.
roman-empire	.3051	.3687	.4123	+6.36	+10.72	1.69 ^a
Penn94	.7649	.8053	.7838	+4.05	+1.89	0.47
chameleon-filtered	.2990	.3707	.3152	+7.17	+1.63 (ns)	–
Actor	.3423	.3667	.3471	+2.43	+0.48 (ns)	–
amazon-ratings	.3679	.3763	.3721	+0.84	+0.41	0.50
squirrel-filtered	.3449	.3357	.3482	−0.91 (ns)	+0.33 (ns)	–
minesweeper [†]	.9059	.9982	.9018	+9.23	−0.41	−0.04
tolokers [†]	.8074	.9202	.8221	+11.28	+1.47	0.13
questions [†]	.6070	.9946	.6759	+38.76	+6.88	0.18

7 Benchmark Granularity and Minimum Detectable Effects

Under our split-level paired design ($n=10$ splits, Penn94 $n=5$), the minimum detectable effect at $\alpha=.05$ and power .8 is $\text{MDE} \approx 3.15 \sigma_d / \sqrt{10}$ (Penn94: $3.72 \sigma_d / \sqrt{5}$), where σ_d is the per-split difference SD. From our runs: Cornell, Texas, and Wisconsin have 37, 37, and 51 test nodes per split, which is 2.70, 2.70, and 1.96 pp per test node, and their oracle-versus-uniform MDEs are 4.1, 4.9, and 4.2 pp. The modern datasets resolve far smaller effects (amazon-ratings 0.37 pp, Penn94 0.38, roman-empire 1.34, chameleon-filtered 2.17). Effects of the size typically reported on WebKB, between one and three points, are therefore at or below the resolution of the standard evaluation design there, before any question of mechanism attribution arises. The full table and per-dataset σ_d appear in App. C.

7.1 Published claims vs. benchmark resolution

Table 7 compares improvements claimed by published methods on the WebKB datasets, computed as each method’s reported accuracy minus the strongest baseline in that paper’s own results table, against the MDE we measure. Of the eleven dataset-level deltas, four are below the MDE (GPR-GNN’s Cornell and Texas margins, and ACM-GNN’s Texas and Wisconsin margins). Two patterns recur. Several flagship deltas depend on which baselines a paper includes: Geom-GCN’s large margins are against GAT, in a table that contains no MLP, while later papers’ own tables show MLP at 91 to 94% on the same datasets. And methods with strong absolute numbers often have WebKB margins near or below resolution: GPR-GNN ties MLP on Cornell, and ACM-GNN’s Texas margin of +2.99 pp carries a reported ± 3.15 SD.

We present this as an audit of benchmark resolution, not of the methods. Most of these papers also evaluate on larger datasets, and none reports a mechanism control of the kind proposed here, which is the gap this paper addresses.

8 Router Diagnostics

Router diagnostics report normalized usage entropy (collapse detection), argmax agreement with the oracle assignment, per-node gate entropy and top-1-minus-top-2 gate margins, and the mutual information of the dominant expert assignment with local homophily, judged against a 200-permutation shuffled null. Single-expert baselines identify which expert carries each dataset, and decision rule D3 closes through the parameter-identical UNIFORM together with the oracle-minus-random contrast (§6) and the frozen-model interventions (§6.1). Full per-dataset diagnostics ship with the release. On the audited diagnostic set: the planted strong-signal profile shows MI effect 0.974 with local homophily (permutation $p=.005$); on real data the dominant-assignment MI is small but decisively nonrandom on Penn94 and roman-empire (effects 0.024 and 0.008, both $p=.005$) and indistinguishable from the null on minesweeper and squirrel-filtered ($p=.27$ and $.07$), with per-node gate entropy around 0.73 to 0.77 and mean top-1 margins of 0.31 to 0.39: soft but genuinely differentiated gates. Permutation p-values are the effect scale of record; the earlier z-score column is retained as a legacy diagnostic only.

9 Discussion

Why can a uniform mixture beat an oracle router? Several explanations are plausible. On tiny graphs, equal mixing acts as an ensemble and a regularizer, while hard routing fragments already-scarce train-

Table 7: Claimed WebKB improvements vs. measured benchmark resolution. Δ is the paper’s reported accuracy minus the strongest baseline in the same paper’s table (baseline named). MDE is from our split-level design (App. C); published evaluations are typically unpaired and have less power. Split protocols differ across papers (noted); Δ s are as-published, not re-run.

Paper	Dataset	Δ (pp)	vs. baseline	MDE (pp)	$\Delta \geq \text{MDE?}$
Geom-GCN ^a (Pei et al., 2020)	Cornell	+6.49	GAT	4.1	yes
	Texas	+9.19	GAT	4.9	yes
	Wisconsin	+14.71	GAT	4.2	yes
GPR-GNN ^b (Chien et al., 2021)	Cornell	+0.00	MLP	4.1	no
	Texas	+0.66	MLP	4.9	no
ACM-GNN ^b (Luan et al., 2022)	Cornell	+4.60	MLP	4.1	yes
	Texas	+2.99	MLP	4.9	no
	Wisconsin	+2.88	MLP	4.2	no
AD-GNN ^c (Hevathige et al., 2026)	Cornell	+12.00	DirGNN	4.1	yes
	Texas	+9.87	GraphSAGE	4.9	yes
	Wisconsin	+12.70	GraphSAGE	4.2	yes

^aGeom-GCN’s table contains no MLP baseline; MLP exceeds all its reported numbers in later papers’ tables.

^bDense 60/20/20 splits. ^cMain table excludes heterophily-specialized baselines (deferred to their supplement).

ing signal across experts. Local-homophily estimates on nodes with three to five neighbors are noisy. And an expert family may simply not span behaviors that differ usefully on a given graph. Our results cannot fully separate these explanations, and we do not attempt to adjudicate among them here. The protocol’s job is to detect that mechanism validation has failed, not to diagnose the architecture.

What should a method paper report? Table 2. The cost is modest, since the oracle, random, and uniform conditions reuse the method’s own training loop. For depth-adaptive methods we additionally suggest an even cheaper pre-test: the ratio of fixed-depth-4 to fixed-depth-2 accuracy, d_4/d_2 , computed from two plain baseline runs. When $d_4/d_2 < 1$, deeper uniform processing already hurts, which is the regime where shallow-biased routing can have value; when $d_4/d_2 \geq 1$, there is little for depth adaptation to protect. It plays the same role for depth that the oracle gap plays for expert choice, as a regime indicator rather than a magnitude predictor. The payoff of the full checklist is the difference between showing that a model scores higher on WebKB and showing that its router demonstrably routes. Algorithm 1 states the full procedure; auditing a new method means identifying its assignment signal, building the four conditions and a mechanism-matched control, and reporting the resulting row.

Cost. A router can also earn its keep through computation rather than accuracy, for example by letting nodes exit early. Our protocol measures signal, not cost, and our recorded wall-clock times measure training through condition-specific early stopping rather than inference latency, so we make no efficiency claims in either direction. For deployment-oriented routers the protocol extends directly by reporting each condition’s accuracy per unit of

inference cost; the falsification logic is unchanged.

What the landscape suggests. The conditional pattern (channel-mixing signal but no topology-expert signal on WebKB, and large headroom for both on modern suites) is consistent with the theory of Wang et al. (2024): which adaptation axis has signal depends on neighborhood distributions and degree structure, not on heterophily as a single scalar. Benchmark builders could report oracle-headroom profiles per mechanism family as part of dataset documentation. Emerging standardized suites are natural substrates for the findings matrix: in particular, GraphBench’s algorithmic-reasoning tasks (Stoll et al., 2025), whose per-node answers have provable computational-depth requirements and whose size-generalization ladder modulates required depth by construction, supply exactly the depth-planted control that §4.4 leaves as the natural extension; and its SAT algorithm-selection tasks connect our D1 to the solver community’s long-standing virtual-best-versus-single-best-solver gap, of which ORACLE-minus-UNIFORM is the per-node analogue.

9.1 When the protocol itself can mislead

The oracle is a measuring device, and like any device it can misread in three ways, all of which appear in our own results.

First, an oracle built on too weak a signal understates headroom. On roman-empire the homophily-bucket oracle gains +12.1 pp while the label-free structural router gains +16.4 pp: the oracle lower-bounds attainable routing given its particular signal, and Proposition 1 explains why a partition-aware field can carry information the homophily buckets cannot.

Second, an oracle built on too informative a signal overstates headroom. On the imbalanced binary

datasets the label-derived oracle approaches the AUC ceiling, because knowing a node’s true local homophily together with its neighborhood strongly constrains its label; those gaps are upper bounds to be read alongside the label-free numbers.

Third, and most extreme, an oracle can be pure label leakage. Define a hindsight oracle that routes each test node to whichever single expert happens to classify it correctly. On Cornell this reaches 82.6% accuracy against 66.9% for uniform, a gap of +15.7 pp ($p_{\text{Holm}} = .0039$ at the split level). This number is not routing signal. The three experts agree on only 62% of test nodes, disagreeing on hard nodes largely at random, and selecting the correct expert per node harvests that randomness using the test label itself; no router without the label could approach it. This is the ensemble-oracle bound familiar from early-exit networks, and it is why oracle choice matters: the oracle must encode a structured, generalizable regime signal, not per-node correctness. Our homophily buckets and planted regime identities are of the first kind; hindsight selection is of the third.

10 Limitations

Our oracle definitions (homophily buckets, with hard and soft variants) are reasonable but not unique, and we swept them: thresholds (0.30, 0.70) and (0.45, 0.55) and the soft mapping leave the WebKB null and the roman-empire positive intact, while minesweeper’s oracle headroom is threshold-sensitive (+7.1 to +9.3 across variants) and squirrel-filtered stays null or negative. The roman-empire result shows that a structural signal can exceed a homophily oracle, and Proposition 1 gives a formal mechanism by which that is possible: no function of degree and homophily reproduces a community-partition-aware field. We have not, however, run the ablation that would attribute the roman-empire result to such a field (the deployed router does not compute R_c), so that attribution remains a consistent hypothesis rather than a demonstrated cause. Oracle gaps are lower bounds on attainable-router headroom given a particular oracle signal. AD-GNN results use our reimplementations from the published formulas, with no mechanism-matched planted control yet (§4.4). Topo-MoE’s expert family is one instance of topology-expert mixtures, and other expert families may find WebKB signal, as ACM-GNN does. The binary-dataset oracle saturation (§6) means those upper bounds are loose, and two questions/UNIFORM runs sit at a constant-prediction collapse boundary where training is chaotically sensitive, a documented instability retained in the variance. The published-deltas audit (Table 7) covers four representative methods; H2GCN and FAGCN rows await verification against their PDFs, and all Δ s are as-published across differing split protocols. Finally, the protocol is an analysis instrument, not a deployment method: the

oracle consumes true labels and cannot run at test time. Its role is to measure whether signal exists, not to route.

11 Conclusion

Per-node adaptive GNNs make a mechanism claim, and mechanism claims need mechanism controls. With a calibrated oracle-versus-uniform protocol we find that routing signal is a property of the (architecture, benchmark) pair. It is undetected for topology-expert mixtures on the WebKB classics, where a label-informed oracle cannot beat no-routing under any setting we swept; it is present for channel mixing on those same graphs; and it is real but only partially realized on modern corrected benchmarks. None of these distinctions is visible in aggregate accuracy. We release the protocol, the planted-regime generators, the full audit trail, and all runs (15,228 training runs in total, including the superseded and quarantined runs retained as the audit trail), and we suggest the checklist of Table 2 as a reporting standard for adaptive-GNN papers.

Reproducibility statement

Every number in this paper is regenerated by script from versioned run CSVs (git tag **results-v2.1**) carrying full provenance stamps (code, configuration, data, initialization, and environment fingerprints); configs pin all seeds, splits, and hyperparameters, and every gate, band, and tolerance is committed before its batch runs; datasets are public (PyG loaders, the Platonov corrected suite, and LINKX Penn94); Penn94’s wide features are reduced by a fixed-seed random projection to 256 dimensions, identically across all arms. Runs execute in a pinned determinism envelope (deterministic kernels, single thread), which makes results bitwise reproducible going forward; the historical runs of the large sparse datasets predate the envelope and reproduce statistically rather than bitwise. A machine-checkable claims ledger binds each number in this paper to a query against the released CSVs, and the release gate refuses claims the data does not support. The protocol package, configs, and run CSVs are public at the repository accompanying this report, together with a contribution guide: the findings matrix of Table 1 is maintained as a living table, and audited rows for additional methods and benchmarks are welcome.

References

- S. Teerapittayanon, B. McDanel, H. T. Kung. BranchyNet: Fast inference via early exiting from deep neural networks. *ICPR*, 2016.
- O. Shchur, M. Mumme, A. Bojchevski, S. Günnemann. Pitfalls of graph neural network evaluation. *Relational Representation Learning Workshop, NeurIPS*, 2018.

- H. Pei, B. Wei, K. Chang, Y. Lei, B. Yang. Geom-GCN: Geometric graph convolutional networks. *ICLR*, 2020.
- J. Zhu, Y. Yan, L. Zhao, M. Heimann, L. Akoglu, D. Koutra. Beyond homophily in graph neural networks. *NeurIPS*, 2020.
- E. Chien, J. Peng, P. Li, O. Milenkovic. Adaptive universal generalized PageRank graph neural network. *ICLR*, 2021.
- D. Lim, F. Hohne, X. Li, S. L. Huang, V. Gupta, O. Bhalerao, S. N. Lim. Large scale learning on non-homophilous graphs. *NeurIPS*, 2021.
- S. Roller, S. Sukhbaatar, A. Szlam, J. Weston. Hash layers for large sparse models. *NeurIPS*, 2021.
- T. Stoll, C. Qian, B. Finkelshtein, A. Parviz, D. Weber, F. Frasca, H. Shavit, A. Siraudin, A. Mielke, M. Anastacio, E. Müller, M. Bechler-Speicher, M. Bronstein, M. Galkin, H. Hoos, M. Niepert, B. Perozzi, J. Tönshoff, C. Morris. Graph-Bench: Next-generation graph learning benchmarking. arXiv:2512.04475, 2025.
- S. Luan, C. Hua, Q. Lu, J. Zhu, M. Zhao, S. Zhang, X.-W. Chang, D. Precup. Revisiting heterophily for graph neural networks. *NeurIPS*, 2022.
- Y. Yan, M. Hashemi, K. Swersky, Y. Yang, D. Koutra. Two sides of the same coin: Heterophily and over-smoothing in graph convolutional neural networks. *ICDM*, 2022.
- N. Dikkala, N. Ghosh, R. Meka, R. Panigrahy, N. Vyas, X. Wang. On the benefits of learning to route in mixture-of-experts models. *EMNLP*, 2023.
- O. Platonov, D. Kuznedelev, M. Diskin, A. Babenko, L. Prokhorenkova. A critical look at the evaluation of GNNs under heterophily: Are we really making progress? *ICLR*, 2023.
- S. Luan, Q. Lu, C. Hua, X. Wang, J. Zhu, X.-W. Chang, G. Wolf, J. Tang. Are heterophily-specific GNNs and homophily metrics really effective? Evaluation pitfalls and new benchmarks. arXiv:2409.05755, 2024.
- J. Wang et al. Understanding heterophily for graph neural networks. *ICML*, 2024.
- K. Wang et al. The heterophilic snowflake hypothesis: Training and empowering GNNs for heterophilic graphs. *KDD*, 2024.
- A. Hevathige et al. Beyond fixed depth: Adaptive graph neural networks for node classification under varying homophily. *AAAI*, 2026. arXiv:2511.06608.
- Y. Abbahaddou, F. D. Malliaros, J. F. Lutzeyer, M. Vazirgiannis. ADMP-GNN: Adaptive depth message passing GNN. *CIKM*, 2025. arXiv:2509.01170.
- J. Boenke et al. Early-exit graph neural networks. arXiv:2505.18088, 2025.
- A. Hevathige, A. N. Zehmakan, Q. Wang. Depth-adaptive graph neural networks via learnable Bakry-Émery curvature. *KDD*, 2025. arXiv:2503.01079.
- S. Holm. A simple sequentially rejective multiple test procedure. *Scand. J. Statist.*, 6(2):65–70, 1979.

A PC-1: the uncalibrated control and per-regime sanity

PC-1 (without M-noise calibration) saturates at $\lambda=1$: every arm reaches accuracy 1.0, so the endpoint gap is zero by ceiling rather than by measurement. This is the reason the calibrated controls exist, and PC-1 is retained precisely to exhibit the artifact. Its historical runs are part of the audit record: the contamination described in App. F touched cells in the historical PC ladders, and PC-1’s role in this paper is therefore served by its clean rerun under the deconfounded generator (the strong-signal profile), whose per-regime instrument checks pass with the raw-feature M-probe at 1.0, by design and gated as such. Historical per-regime accuracy at $\lambda=0$ (mean over 30 runs; the capacity-UNIFORM row belongs to the superseded smaller-uniform design):

Condition	L-regime	H-regime	M-regime
ORACLE	.871	.644	1.000
LABEL-FREE	.887	.604	1.000
UNIFORM	.337	.521	1.000
RANDOM	.489	.446	.678
single-L	.841	.541	.205
single-H	.316	.677	1.000
single-M	.338	.519	1.000
capacity-UNIFORM	.336	.512	1.000

B TopoMoE development history (Week-1 gate and baseline matrix)

The pre-registered Week-1 gate (beat gated-GAT on at least 4 of 6 heterophilic datasets) returned 3 of 6: wins on Cornell, Texas, and Wisconsin (+22.7, +17.5, +19.6pp, all $p < 10^{-9}$), losses on Chameleon and Squirrel, and no significant difference on Actor. In the 10-model strong-baseline matrix (GCN, GAT, GatedGAT, FAGCN, GPRGNN, H2GCN, LINKX, GNN-MoE, ACMGNN, TopoMoE), TopoMoE ranked 1st on Texas, 3rd on Cornell, 5th on Wisconsin, 4th on Squirrel, and 6th on Chameleon. GatedGAT sits in the bottom quartile on the WebKB trio, which is what made the Week-1 wins misleading. Full tables ship with the release. This history motivated the protocol, and we include it for transparency.

C MDE table

Per-dataset paired-difference SDs and MDEs (split-level design, $n=10$ splits; $\text{MDE} = (t_{.975,9} + t_{.80,9}) \sigma_d / \sqrt{10}$; Penn94 uses 5 splits, $\text{MDE} = (t_{.975,4} + t_{.80,4}) \sigma_d / \sqrt{5}$): Cornell $\sigma_d=.041$, MDE 4.1 pp; Texas .049, 4.9; Wisconsin .042, 4.2; Actor .0082, 0.81; amazon-ratings .0037, 0.37; Penn94 .0023, 0.38; roman-empire .013, 1.34; chameleon-filtered .022, 2.17; squirrel-filtered .016, 1.59. Accuracy granularity per single test node: Cornell and Texas 2.70 pp, Wisconsin 1.96 pp.

D Hyperparameters and configs

All configs (architecture widths, depth levels, the initial-residual coefficient, stochastic-depth and self-distillation weights, optimizer settings, split seeds, oracle bucket thresholds, ACM soft-oracle weights, and the AD-GNN reimplementations constants) are pinned in the versioned `configs/` directory of the protocol package (git tag `results-v1`) and are available with the release. Key defaults: hidden width 64, depth levels $\{2, \dots, 6\}$, initial-residual $\alpha=0.15$, stochastic-depth $p=0.10$, self-distillation weight 0.10, Adam (10^{-2} , weight decay $5 \cdot 10^{-4}$), 200 epochs with validation-selected checkpoints, oracle homophily buckets at (0.4, 0.6), and ACM soft-oracle weights (0.8, 0.1, 0.1).

E A/A calibration of the statistical pipeline

As a check on the statistics themselves, we ran the identical paired machinery on A/A comparisons: for each router condition, dataset, and pair of seeds, we tested the condition against itself, paired by split (three seed pairs per cell). Under a true null this should produce Holm-significant results at roughly the nominal rate. It did not: of 387 such comparisons, 46 had a raw Wilcoxon $p < .05$ (a 95% binomial interval for a true 5% rate is 11 to 28), and 12 survived Holm correction. The failures were not scattered noise but a consistent pattern. On roman-empire’s label-free router, for example, every one of the three seed-pair comparisons was Holm-significant with a transitive quality ordering: seed 0 beats seed 1 by +1.32 pp, seed 0 beats seed 2 by +2.04 pp, and seed 1 beats seed 2 by +0.72 pp, consistent across splits. The cause is that `run_one` sets the random seed before model construction, so for a fixed dataset a given seed reproduces the identical weight initialization on every split; some initializations are simply better than others, making seeds technical replicates rather than independent samples. The remedy is the split-level design used throughout this paper: average seeds within each split first, then treat the resulting per-split differences (10 per dataset, 5 for Penn94) as the paired sample. This is the paper’s own calibration philosophy, applied to its own

statistics: an instrument is only as trustworthy as the negative control run against it.

F The audit trail: how this paper’s evidence was corrected

This paper’s thesis is that mechanism claims need mechanism controls. Between its first draft and this version, that thesis was applied to the paper itself, through an external code-level review and an internal integrity system built in response. We summarize the findings here because several corrected numbers in the main text are only interpretable against this record, and because the corrections are, collectively, the strongest evidence the paper offers that its protocol matters.

The capacity confound. The review found that our original UNIFORM condition on the planted controls and modern benchmarks was not the routed architecture with equal gates but a smaller shared-stack model with roughly one third of the parameters, contradicting the paper’s own same-architecture requirement. Rerunning UNIFORM as the parameter-identical equal-gate model changed real conclusions: Penn94’s label-free recovery fell from 0.81 to 0.47, and minesweeper’s apparent +9.4 AUC-point label-free gain reversed to -0.41 ; it had been model size. The WebKB experiments were unaffected (their uniform was always the same architecture). The old shared-stack model is retained in the release as a secondary baseline under the name `shared_mean`.

The generator confound and the cache incident. The original dilution generator coupled feature noise to λ ; the deconfounded generator (PC-1c) varies regime assignment only, under nested conversions and held-out seeds. Its first run appeared to FAIL its acceptance gate, and the investigation found the recorded rows were fossils: the experiment runner’s resume cache identified results by name rather than content, so rows computed by an intermediate code state survived later fixes. Every result row now carries content fingerprints (code, config, data, initialization, environment), and a cached row is reused only when the data it was computed from hashes identically today.

The endpoint gate and its correction. The clean PC-1c then genuinely failed its preregistered $\lambda=1$ null, and the failure was diagnostic: at $\lambda=1$ the oracle degenerates to a constant all-M policy, so oracle-minus-uniform measures a homogeneous expert-composition residual rather than per-node routing (the usage-matched random condition coincides with the same constant policy there, gap exactly zero). The corrected endpoint null, oracle-minus-best-single, was preregistered for a replication on ten fresh generator seeds and passed (Table 3).

Confirmed contamination and its replacement.

A quantitative signature screen over all historical families then found real corruption: whole split-blocks of adjacency-dependent arms collapsed to chance in the historical PC-1b ladder and in the published-methods planted rows, caused by a process-global adjacency cache keyed on a reusable memory address, so that a later same-shape graph could silently receive an earlier graph’s adjacency. Isolated reruns confirmed contamination cell by cell (a poisoned run cannot reproduce in a single-graph process; 88 of 94 flagged cells came back healthy). All affected families were replaced by clean reruns under the integrity harness, which is where the corrected planted-control values in §4 and Table 5 come from. The same screen and rerun protocol EXONERATED the WebKB evidence: the clean rerun reproduces every one of the 27 accepted WebKB group means to four decimal places. The same screen and rerun protocol produced both verdicts, the condemnation and the exoneration, and that is why we trust either one. Separately, the paper’s central WebKB result (Table 4), whose original runs were imported from an earlier harness outside this repository’s audit scope, was re-derived from scratch in the audited harness: every conclusion reproduces, all three oracle-minus-uniform contrasts are nulls at $p_{\text{Holm}}=1.0$, and the cross-harness point-estimate drift (up to about 3 pp on the uniform arm, more on the freshly drawn random assignments) changes no verdict. One null cell’s point estimate crossed zero in the re-derivation, which is what nulls do across independent harnesses.

The standing guards. The release now runs under a written integrity standard: provenance stamps on every row; content-keyed caching; pre-training instrument gates on every generated graph (exact wiring fractions, planted-signal probes, nesting and single-variable invariants); post-training sanity flags (chance collapse, ceiling, majority collapse, undertraining, shared initialization) with an explicit explanation allowlist; runtime parameter-identity and router-freeze assertions; a pinned determinism envelope; declared statistical families with a preregistered primary test; and a machine-checkable claims ledger binding the paper’s numbers to the released CSVs. Seven incidents are on the public register (I-13 through I-19), six of them errors in our own gate specifications that the system forced us to correct and one confirmed data contamination, and each ruling that dismissed an alarm or upheld one is part of the released record: the stop reports, rulings, signature screens, cell-level confirmation reruns, coverage audits, and the release gate’s verbatim verdicts all ship under **results/integrity/** in the repository, with every artifact SHA-256 inventoried.